

Impact of Artificial Intelligence Generated Feedback on Math Quiz Scores

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Abstract

This study examined the effect of feedback on students' responses to quiz questions generated by artificial intelligence (AI), specifically using ChatGPT-3.5, on student math quiz performance at a private school in the Philippines. Immediate, personalized feedback is known to support effective learning, yet teachers often struggle to provide it consistently due to large class sizes and limited teaching time. Over 14 weeks, 170 students from Grades 1 to 10 participated in this study, completing 1,348 quiz response sets using the school's existing online learning system, the Moodle Learning Management System (LMS). Students were randomly assigned to two groups: one group received basic correctness-only feedback, while the other received detailed, AI-generated feedback that included explanations, hints, and encouragement tailored to their responses. Students scoring below 80% on initial quizzes received feedback from their group before retaking similar quizzes. Those who received detailed AI-generated feedback improved significantly more ($\sigma = 1.1$, Hake's $g = 21.6\%$) than the correctness-only group ($\sigma = 0.7$, Hake's $g = 13.8\%$), a statistically significant difference ($p < .01$, Cohen's $d = 0.40$). These results indicate that detailed AI-generated feedback effectively supports math learning, providing scalable, personalized instructional assistance within everyday classroom environments across diverse grade levels.

1. Introduction

The use of Artificial Intelligence (AI) is undeniably becoming popular worldwide. Many businesses have been using it in various forms for decades already. Now, it is being adopted rapidly in education. Among AI technologies, large language models (LLMs) are particularly promising. These LLM-based tools can function like personal tutors, offering lessons, practice exercises, clear explanations, hints, and encouragement. Students already use such AI tools for homework, understanding complicated topics, or summarizing readings (Pardos & Bhandari, 2024). Educational research emphasizes that timely feedback matters. Students learn more effectively when they receive feedback immediately after making a mistake (Shute, 2008). Since AI tutors can respond instantly, they may significantly help student learning.

However, providing timely and individualized feedback is often challenging for teachers in regular classrooms. Large class sizes and limited instructional time make one-on-one teaching challenging (Paris, 2022). Bloom (1984) posted this as the two-sigma challenge in his study, where he reported that students with personal tutors scored about two standard deviations higher than students in typical classes. This suggests that AI could be the solution to the challenge by providing valuable personalized feedback on a larger scale.

Students are already frequently using AI tools for educational purposes. A recent study reported that by 2025, around 26% of U.S. teens had used ChatGPT for school-related tasks, a doubling from 2023 (Sidoti et al., 2025). Even being highly popular among students, approximately 25% of teachers remain skeptical. A significant portion of this group believes that AI may cause more harm than good for learners (Lin, 2024). The mixed opinions call for a better understanding of how to use and integrate AI into classrooms effectively and to avoid potential pitfalls

The teachers' skepticism is not unfounded. Recent research warns that AI may negatively impact learning if it provides students with answers too readily. Bastani et al. (2024) studied high school algebra students who used GPT-4 to receive complete solutions to problems. Initially, these students performed well, but later their performance dropped by around 17% compared to students who learned without AI. However, students who received only hints from AI did better. Giving complete answers too readily might reduce student engagement in problem-solving (Bastani et al., 2024).

Other studies present more positive findings when AI provides guidance instead of just answers. Pardos and Bhandari (2024) compared AI-generated hints with those from human tutors in math problems, finding students learned equally well from both. Steiss et al. (2024) also observed that although teacher feedback was generally clearer, AI was good in providing consistent feedback on how to follow writing rules and guidelines. Poláková and Ivenz (2024) support this when they found that students improved their essays after receiving feedback generated by AI. The students described the feedback as being clear and helpful.

Overall, existing research indicates that AI-based feedback can be beneficial for students. The current challenge is to assess its various levels of effectiveness based on how it is used. If AI provides thoughtful hints and explanations instead of simply giving answers, it can offer learning support comparable to human guidance. If AI only provides answers without engaging students, it may hinder understanding. Most studies have involved older learners (high school or college students) in controlled settings. There is limited knowledge on how younger students in regular classroom environments respond to AI tutoring. Furthermore, it remains unclear how detailed AI-generated feedback compares to basic correctness-only feedback, commonly used by teachers.

The current study addresses these understanding limitations on how younger learners benefit from AI by investigating the impact of an AI tutoring system in real classroom conditions across multiple grade levels. Specifically, we examined how AI-generated feedback, using ChatGPT-3.5 as a virtual tutor within the school's existing Moodle Learning Management System (LMS), influences student learning in mathematics. The study included students from Grades 1 to 10 at a private school in the Philippines. Our primary aim was to compare detailed, explanatory AI-generated feedback directly against simple correctness-only feedback provided by teachers, assessing its impact on student quiz performance.

This research contributes to the field by exploring AI tutoring across younger age groups (Grades 1–10), utilizing authentic classroom settings without researcher interference, and demonstrating the practical application of existing Moodle LMS and school technology. The findings of this study provide important insights into whether detailed AI-generated feedback can meaningfully enhance student learning compared to typical teacher feedback practices.

2. Materials and Methods

Before the study, the researchers obtained approval from the institution's Ethics Committee. They also obtained written permission from the school's director, based on informed consent from participating students and their guardians.

The study used the standard randomized alternate-treatment design with matched pairs. A total of 248 students from Grades 1 to 10 joined the study. Students were paired by grade based on their scores in a diagnostic math test. In each pair, one student was randomly placed in the AI feedback group (Group AI). The other student was placed in the correct-answer-only group (Group R/W).

The study lasted 14 weeks. Students worked through several math topic modules. These modules were part of their normal class routine. The researchers helped with setup, training, and orientation. After that, the school team managed the sessions. Researchers did not intervene once the study began. Some sessions were done at home due to high temperatures. Others were affected by school holidays. These were treated as part of normal school conditions.

The interactive lessons, exercises, and quizzes for this study were delivered to the student tablets using Moodle, the school's learning management system. The study followed a mastery learning approach based on Bloom's model. Each topic module included a short lesson, an exercise with unlimited tries, and three quizzes.

As shown in Fig. 1, each module started with a lesson and an exercise set. After that, students took Quiz 1. This quiz consisted of five multiple-choice questions and was used to assess their understanding. If a student scored 80% or higher, they skipped to Quiz 3. If they scored below 80%, they took the one-question Quiz 2.

Quiz 2 was used to give feedback. The same question was shown to both groups. One group received feedback from ChatGPT-3.5. This feedback included step-by-step explanations and a short message of encouragement. The other group saw only the correct answer, with no explanation. The prompt sent to the AI included the question, all answer choices, the correct answer, and the student's chosen answer.

After Quiz 2, all students took Quiz 3. This quiz had five new questions about the same topic. After Quiz 3, all students received AI-generated feedback, regardless of group. This step helped ensure that everyone had the same experience before moving on to the next topic.

Before the study began, the researchers reviewed examples of AI feedback. Ten samples were reviewed for three randomly chosen questions per grade. A researcher with training in math education checked the feedback. The review showed that the AI feedback was accurate, clear, and encouraging.

All quiz scores and activity logs were recorded through Moodle. Student names were removed and replaced with anonymous ID codes. Only students who completed all three quizzes in a module were included in the analysis. For the main results, only students who completed at least three full modules with feedback were included in the analysis. For topic-level results, only modules with at least five students in each group were included.

In total, 170 students met these requirements. Across all topics, the study collected 1,348 three-quiz response sets that included feedback.

Scores needed in the study were recorded at three points per module: Quiz 1 (before the feedback), Quiz 2 (during the feedback for those who got less than 80% from Quiz 1), and Quiz 3 (after the feedback). The primary outcome measure was improvement from Quiz 1 to Quiz 3.

The primary outcome was the improvement in students' performance from Quiz 1 to Quiz 3. The effectiveness of feedback was evaluated using the following metrics:

The average score improvement is calculated by subtracting the student's Quiz 1 score from their Quiz 3 score. The total improvement in all qualified quiz pairs (Quiz 1 and Quiz 3) scores is divided by the total number of qualified, completed attempts for Quizzes 1, 2, and 3 sets (Bloom, 1984).

Sigma improvement is calculated by normalizing the average score improvement (Quiz 3 minus Quiz 1) by the standard deviation of the Quiz 1 scores for each group. This measure indicates the extent to which the scores have improved relative to the initial score variability (Bloom, 1984).

Consistency of improvements is calculated using the standard deviation of the score improvements (Quiz 3 scores minus Quiz 1 scores) among students for each group. This metric indicates how consistently the feedback improved student scores across each group (Hattie & Timperley, 2007).

Hake's normalized learning gain (g) is determined by dividing the actual improvement between Quiz 1 and Quiz 3 by the maximum possible improvement, which is the difference between a perfect score and the Quiz 1 score (Hake, 1998).

Statistical significance is calculated using a t-test to check whether the differences in score improvements between the two groups are not due to random chance.

Cohen's effect size (d) is computed by getting the difference between the average improvement from AI feedback and from answer-only feedback, and then dividing by the weighted average of the two variances of scores in Quiz 1 (Cohen, 1988).

3. Results

The field study took place in a private school in the Philippines. Grades 1 to 10 students were invited to participate in the study. The total number of participants was 248 students. Of these, only 170 students met the minimum three completed module requirements. Many students, specifically from Grades 2 and 3, got at least 80% in Quiz 1 for the different topic modules and moved straight to Quiz 3 of the same topic module. The filtered 170 students completed Quiz 1, qualified to take and completed Quiz 2, and then completed Quiz 3. These students provided 1,348 sets of valid data points for analysis.

Group R/W had 90 students with 747 sets of three quiz scores. Group AI had 80 students with 601 sets of three quiz scores. In addition, 51 topic modules met the minimum of five participants required for module-level analysis.

Group AI received AI-generated feedback that included explanations, solutions, tips, and encouragement. Group R/W received the correct or incorrect answer as feedback. The schoolwide results showed that Group

AI had significantly greater improvement in scores than Group R/W.

Figure 2 shows the average increase in quiz scores before and after AI feedback. The average score improved from 41.8–54.4%, an increase of 12.6%. This is equal to a 1.1-sigma improvement. The standard deviation of score changes was 11.8%. Hake's normalized gain was 21.7%.

Figure 3 shows the average increase in quiz scores before and after correct-answer feedback for Group R/W. The average score improved from 42.5–50.4%, an increase of 7.9%. This equals a 0.7-sigma improvement. The standard deviation of improvements was 12.7%. Hake's normalized gain was 13.8%.

Table 1
Summary of Results for Schoolwide, Elementary, and High School *The table shows the performance of 170 students across 1,348 three-quiz sets from Grades 1 to 10 and was grouped as Schoolwide, Elementary, and High School.*

Aggregation	Schoolwide		Elementary		High School	
Feedback Groups	AI	R/W	AI	R/W	AI	R/W
Average Score Increase (%)	12.6	7.9	13.6	7.3	11.6	8.5
Std. Dev. of Increases (%)	11.8	12.7	11.1	12.0	12.5	13.5
Increase in Std. Dev. (Sigma)	1.1	0.7	1.2	0.6	1.1	0.8
Hake's Normalized Gain (%)	21.6	13.8	21.8	12.0	21.5	15.6
Cohen's Effect Size (d)	0.4 for AI		0.6 for AI		0.2 for AI	
Significance of Increase (p-value)	0.01 for AI		0.01 for AI		0.14 for AI	

The summary results for the schoolwide, elementary grades (1 to 6), and high school grades (7 to 10) are shown in Table 1.

This difference between the improvement of Group AI and the improvement of Group R/W was statistically significant. The p-value is 0.01, and the effect size was Cohen's d = 0.4, which is close to a moderate effect in favor of the Group AI.

In the elementary grades, Group AI improved from 37.6–51.2%. This is a 1.2-sigma gain and a Hake's gain of 21.8%. Group R/W improved from 39.3–46.6%, a 0.6-sigma gain and Hake's gain of 12.0%. The effect size was 0.6, and the difference was statistically significant ($p < 0.01$).

In the high school grades, Group AI improved from 45.9–57.5% (a 1.1-sigma gain, Hake's gain of 21.5%). Group R/W improved from 45.4–53.9% (a 0.8-sigma gain, Hake's gain of 15.6%). The effect size was 0.2, and the p-value was 0.14, which is not statistically significant.

Table 2

Summary of Results for Grades 1 to 10 The table shows the Quiz score performance per grade level for both Groups AI and R/W. Grades 1 and 2 do not have sufficient data for analysis.

Metrics	Ave. Score Increases (%)		Increases in Std. Dev. (Sigma)		Std. Dev. of Increases (%)		Hake's Normalized Gain (%)		Cohen's Effect Size (d)	Significance of Increases (p-value)
Groups	AI	R/W	AI	R/W	AI	R/W	AI	R/W	AI or R/W	AI or R/W
Grade 1	16.9	9.0	1.3	1.2	12.0	6.3	23.3	24.9	0.7 (AI)	0.06 (AI)
Grade 2	<i>Insufficient data. Most students got at least an 80% score in Quiz 1 for the different topic modules, and there was no need to take Quiz 2 for the differentiated feedback.</i>									
Grade 3										
Grade 4	15.6	5.8	1.3	0.6	11.0	13.7	25.7	10.4	0.9 (AI)	0.06 (AI)
Grade 5	10.7	8.7	1.2	1.0	11.5	11.3	17.3	13.7	0.2 (AI)	0.33 (AI)
Grade 6	11.2	1.0	1.3	0.1	11.1	11.2	19.7	1.7	0.9 (AI)	0.05 (AI)
Grade 7	7.5	13.0	0.5	0.8	14.3	13.6	16.6	24.2	0.4 (R/W)	0.23 (R/W)
Grade 8	9.2	9.7	1.6	1.0	9.5	10.8	16.3	17.8	0.1 (R/W)	0.45 (R/W)
Grade 9	10.8	(0.9)	0.9	(0.1)	11.3	8.8	19.1	(1.6)	1.0 (AI)	0.00 (AI)
Grade 10	18.0	19.2	1.7	2.0	15.5	13.3	33.6	35.0	0.1 (R/W)	0.42 (R/W)

Table 2 shows the performance summary for each grade level. The AI feedback group outperformed the R/W group in most grades, especially Grades 6 and 9.

Statistically significant improvements for Group AI were found in Grade 6 ($p = 0.05$) and Grade 9 ($p = 0.002$). Near-significant improvements were observed in Grade 1 ($p = 0.06$) and Grade 4 ($p = 0.06$). In Grades 2 and 3, most students scored above 80% on the first quiz and skipped the feedback stage; therefore, there was not enough data to analyze those grades.

Group R/W showed slightly better or similar results in Grades 7, 8, and 10, but the differences were not statistically significant.

For module-level analysis, there is a different acceptance criterion to avoid outliers from significantly affecting the group score. Each module should have at least five students per group who received feedback (scoring less than 80% in Quiz 1). With this criterion, only 51 modules were eligible for module-level analysis: 3 from Grade 1, 1 from Grade 4, 10 from Grade 5, 3 from Grade 6, 4 from Grade 7, 13 from Grade 8, 13 from Grade 9, and 4 from Grade 10.

As tabulated in Table 3, of the 51 eligible modules, five modules showed statistically significant results ($p \leq 0.05$), and five more showed near-significant results ($p < 0.10$).

Among the significant results, three modules from Grade 9 favored Group AI, while one module from Grade 5 and one from Grade 8 favored Group R/W. Among the near-significant results, all five modules favored Group AI (2 in Grade 5, 2 in Grade 8, and 1 in Grade 9). The average effect size across these 10 modules was Cohen's $d = 0.6$, with an average Hake's gain of 23.3% for Group AI and 6.5% for Group R/W.

Table 3

Selected Results at Module Level *The table shows the performance metrics for 10 selected topic modules that showed significant or near-significant results. The 10 modules were labeled as Grade Level and Module Quiz number.*

Metrics	Ave. Score Increases (%)		Increases in Std. Dev. (Sigma)		Std. Dev. of Increases (%)		Hake's Normalized Gain (%)		Cohen's Effect Size (d)	Significance of Increases (p-value)
Modules	AI	R/W	AI	R/W	AI	R/W	AI	R/W	AI or R/W	AI or R/W
Gr5, Qz77	(12.0)	13.3	(0.9)	0.8	30.1	28.3	(22.2)	20.7	0.8 (R/W)	0.04 (R/W)
Gr8, Qz40	(16.0)	16.0	(1.8)	0.9	32.9	16.7	(36.4)	30.8	1.0 (R/W)	0.04 (R/W)
Gr9, Qz35	16.0	(5.0)	0.9	(0.3)	16.7	21.1	30.8	(9.7)	1.3 (AI)	0.03 (AI)
Gr9, Qz45	15.6	(4.4)	1.5	(0.4)	19.4	21.9	30.4	(7.7)	1.0 (AI)	0.03 (AI)
Gr9, Qz46	22.9	(16.7)	1.2	(0.8)	33.5	23.4	50.0	(29.4)	1.2 (AI)	0.02 (AI)
Gr5, Qz60	16.0	0.0	0.8	0.0	18.4	26.2	21.6	0.0	0.9 (AI)	0.07 (AI)
Gr5, Qz76	38.2	28.9	2.1	1.4	14.0	14.5	70.0	59.1	0.7 (AI)	0.08 (AI)
Gr8, Qz44	15.0	0.0	1.2	0.0	17.7	23.1	26.1	0.0	0.8 (AI)	0.09 (AI)
Gr8, Qz47	24.0	2.5	1.4	0.1	21.9	22.5	31.6	4.8	1.0 (AI)	0.06 (AI)
Gr9, Qz38	17.1	(2.0)	0.8	(0.1)	21.4	25.7	31.6	(4.0)	0.9 (AI)	0.06 (AI)
Average	13.7	3.3	0.7	0.2	22.6	22.3	23.3	6.5	0.6 (AI)	0.05 (AI)

4. Discussion

This study tested whether AI-generated feedback from ChatGPT-3.5 could help students improve math quiz scores. The results supported our expectation that detailed AI-generated feedback helps students more than just knowing the correct answer. This positive finding aligns with earlier studies that demonstrate the importance of detailed and timely feedback in supporting learning (Shute, 2008; Bloom, 1984).

Younger students (Grades 1–6) benefited the most from AI feedback. Older students (Grades 7–10) also improved, but the improvement was minor and not statistically significant. Younger students may

understand and use detailed feedback better, making it more helpful for them. This finding supports earlier research showing AI feedback can effectively help students learn math (Pardos & Bhandari, 2024).

Comparing these results to earlier studies, our findings match those by Pardos and Bhandari (2024), Steiss et al. (2024), and Poláková and Ivenz (2024). These studies also found positive impacts of detailed AI feedback. However, our results differ from Bastani et al. (2024), who warned that students might learn less if AI gives answers without requiring active thinking. In our study, AI feedback gave clear explanations and encouragement, which likely kept students actively engaged.

Results at the module level further showed that AI feedback was instrumental in specific grades and topics. For example, Grades 6 and 9 had significant improvements. These differences suggest that AI feedback may work best for specific age groups or for challenging math topics.

A strength of this study was that it took place in real classrooms during regular school activities. Teachers used the school's own Moodle LMS and regular devices without special preparation or researcher interference. Therefore, the findings can be easily applied in other schools using similar tools.

The study had several limitations. It was conducted in just one private school, so results may not apply to other contexts or public schools. Data for Grades 2 and 3 were limited because many students initially scored above 80%, so no detailed feedback was required. Also, we compared AI feedback only with correctness-only feedback, not with feedback from human teachers. Other uncontrolled factors, such as student motivation, teacher quality, and classroom conditions, might have affected the outcomes.

Future research should compare AI-generated feedback directly with teacher feedback. It should also test AI feedback in different types of schools and other subjects like Science, History, and Language Arts. Future studies could also investigate different types of assessments, such as open-ended questions or writing assignments. These studies will provide a better understanding to help identify when AI-generated feedback is most helpful and how well it can support students in different contexts.

5. Conclusion

In this study, students who received detailed feedback from ChatGPT-3.5 improved their math quiz scores more than those who only learned the correct answer. Younger students, particularly those in elementary grades, showed the most significant improvement. The results suggest that AI-generated feedback can effectively provide timely and personalized support to students.

The study demonstrates that AI-generated feedback can fit easily into regular classroom teaching using existing tools like Moodle LMS and standard devices. This practical approach could reduce teacher workload by providing helpful feedback to many students simultaneously.

However, since this study took place in a single private school, further research is needed. Future studies should test AI feedback in different schools, subjects, and types of assessments. Such research will help confirm the effectiveness of AI-generated feedback and support broader educational use.

Declarations

The authors did not receive support from any organization for the submitted work.

The authors have no relevant financial or non-financial interests to disclose.

Author Contribution

LRDV, CMA, CPM, EFTL, and DSYT contributed to the research conceptualization and methodology. LRDV did field experiments and data collection. LRDV, CMA, CPM, EFTL, and DSYT performed data analysis. LRDV initially wrote the manuscript. CMA, CPM, EFTL, and DSYT provided substantial comments on how to improve all manuscript versions. LRDV, CMA, and CPM provided the final revision of the manuscript. All authors read and approved the final manuscript.

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Data Availability

Data is provided within the supplementary information files.

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Figures

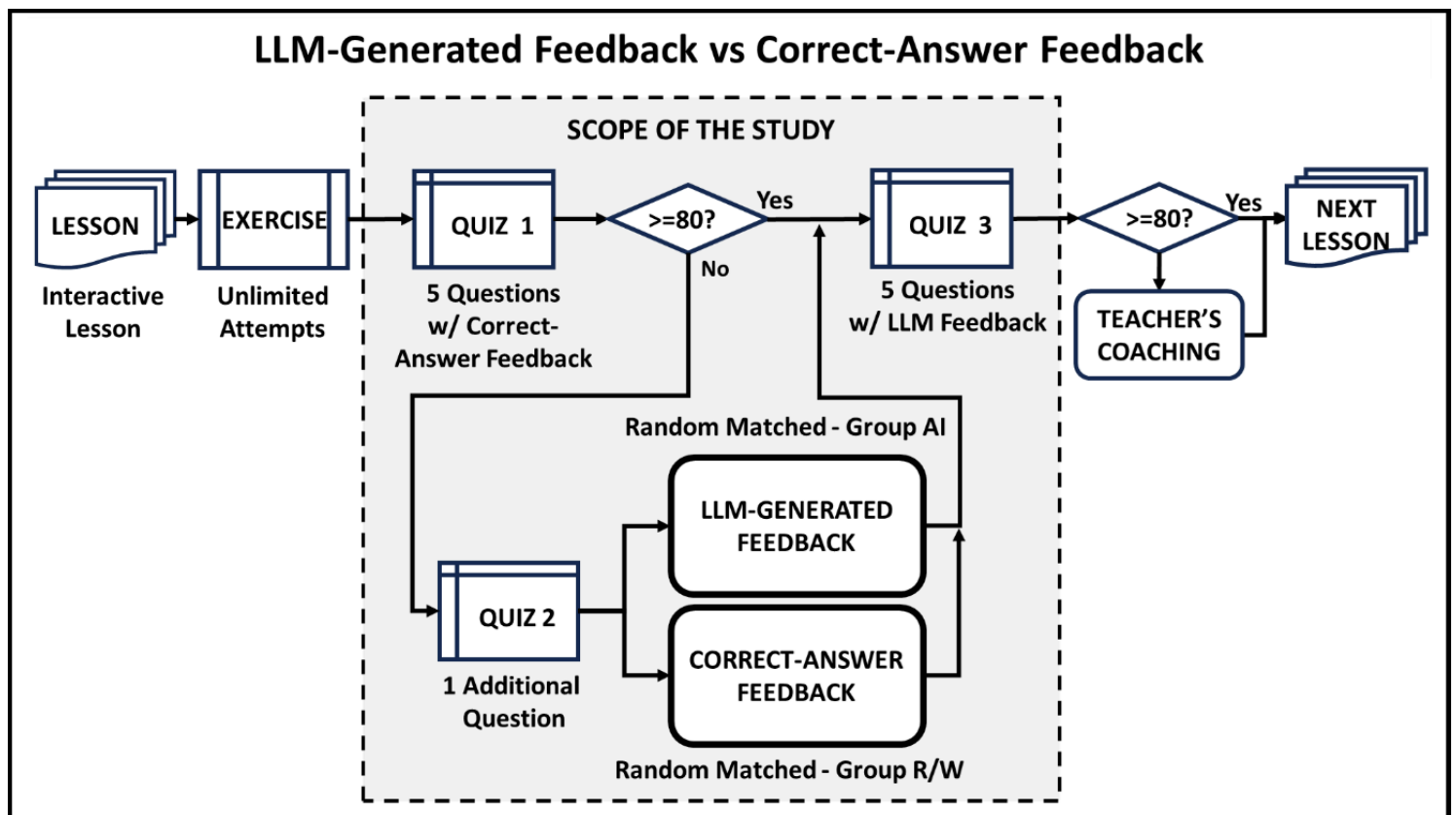


Figure 1

Learning, Assessment, and Feedback Process.

This figure shows the assessment method used to compare AI feedback (Group AI) to traditional correct-answer feedback (Group R/W).

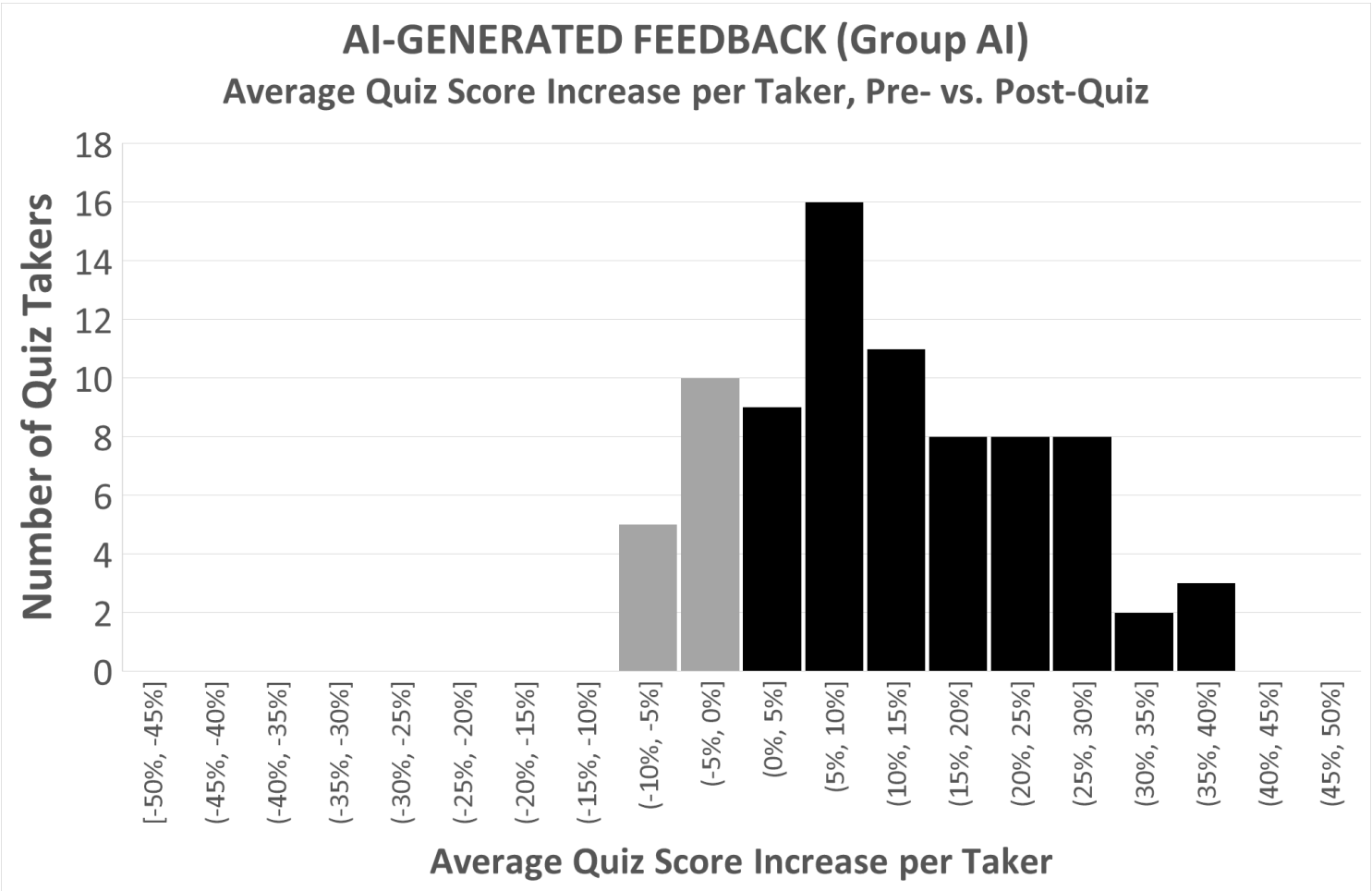


Figure 2

Increase in Average Quiz Scores After AI Feedback

The figure shows the improvement in Math quiz scores for Grades 1-10 students in Group AI before and after receiving AI feedback. The improvements were 1.1 standard deviation (or 1.1-sigma) and 21.7% Hake's normalized gain.

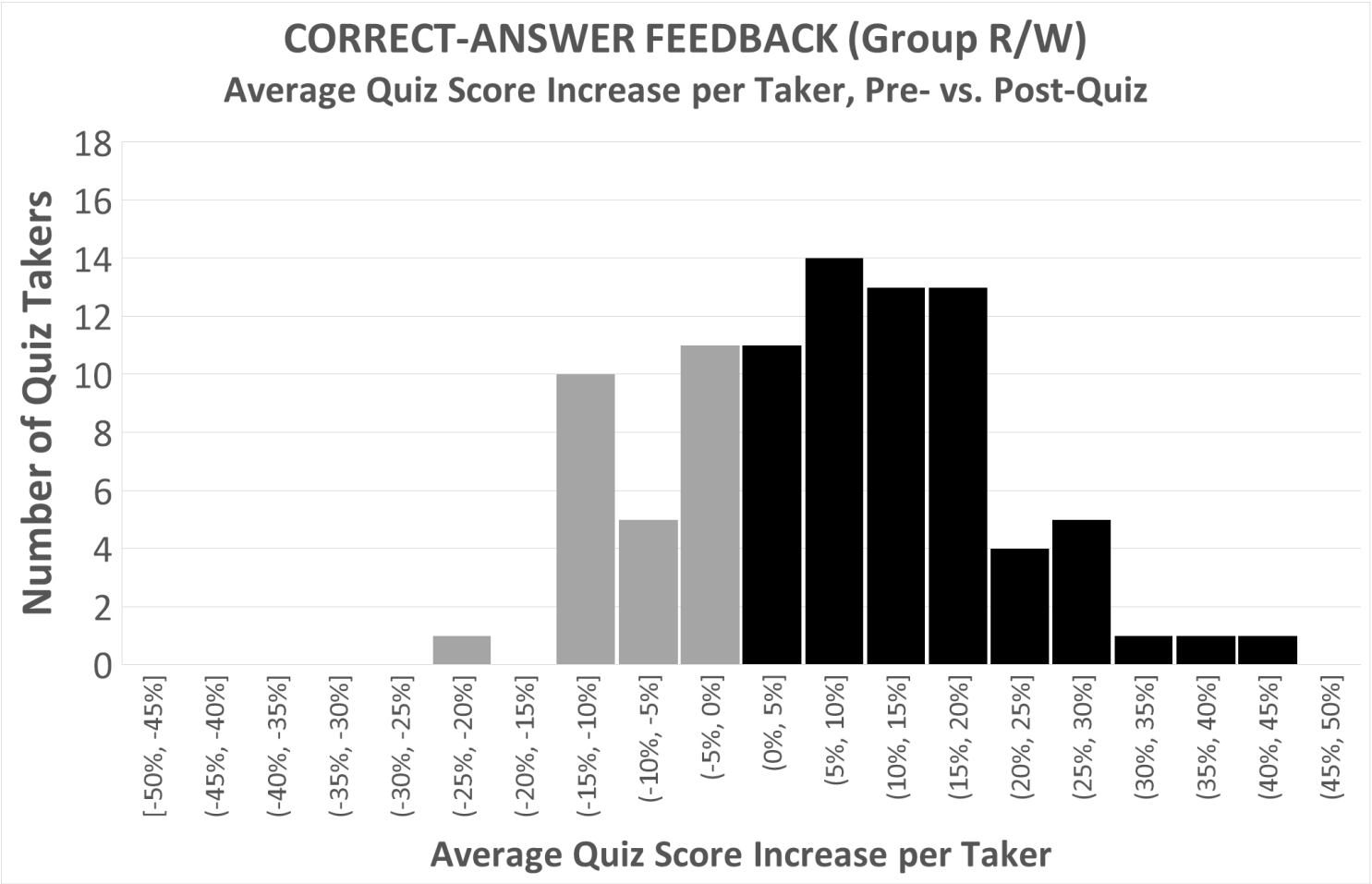


Figure 3

Increase in Average Quiz Scores After Correct-Answer Feedback

The figure shows that Math quiz scores improved for Grades 1-10 students in Group R/W before and after the correct-answer feedback. The improvements were 0.7 standard deviation (or 0.7-sigma) and 13.8% Hake's normalized gain.

Supplementary Files

This is a list of supplementary files associated with this preprint. Click to download.

- [ImpactofAIGeneratedFeedbackonMathQuizScoresData.xlsx](#)